

Learning the Dirichlet–Neumann Operator with Fourier Neural Operators

Depth-conditioned surrogate modeling for nonlinear water waves

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Problem Setup

The Dirichlet–Neumann Operator

The free-surface Euler equations in Zakharov–Craig–Sulem form:

$$\eta_t = \mathcal{G}(\eta; h) \xi, \quad (1)$$

$$\xi_t = -g \eta - \frac{1}{2} \xi_x^2 + \frac{(\mathcal{G}(\eta; h) \xi + \eta_x \xi_x)^2}{2(1 + \eta_x^2)}, \quad (2)$$

where $\eta(x, t)$ is surface elevation, $\xi(x, t)$ is surface velocity potential, h is depth.

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Computational bottleneck: Evaluating $\mathcal{G}$ requires solving Laplace's equation in the fluid domain at every time step. We compute it via the Craig–Sulem series:

$$\mathcal{G}(\eta; h) \xi = \sum_{m=0}^M G_m(\eta) \xi,$$

truncated at order $M = 6$, with $8\times$ -padded FFTs to avoid aliasing.

Goal

Replace the expensive $\mathcal{G}(\eta; h) \xi$ evaluation with a neural surrogate:

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Requirements:

- Accurate across wave regimes: solitons, Stokes waves, random sea states
- Generalize across depths: shallow water ($h = 1$) through deep water ($h = 1000$)
- Stable when plugged into a time integrator for long rollouts ($T \gg 100$)

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Approach:

1. Generate training data via high-fidelity numerical solvers
2. Train a depth-conditioned FNO on $(\eta, \xi, h) \mapsto \mathcal{G}(\eta; h) \xi$
3. Exploit physical structure: learn only the **nonlinear residual** beyond the linear DNO

Numerical Solver

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1. Conformally map to hodograph coordinates $\gamma \in [-s_{\max}, s_{\max}]$ with the transform $\phi(\gamma) = \alpha\gamma + \gamma^p$ ($\alpha = 0.01$, $p = 5$)
2. Iterate fixed-point equations for (τ, θ, F^2) using a strip Hilbert transform (80 iterations per bisection step, 24 bisection steps to hit target amplitude)
3. Recover the physical profile $\eta(x)$ and velocity potential $\xi(x)$ on a periodic grid ($N_x = 1024$, $L = 164$); recompute $\mathcal{G}(\eta; h)\xi$ via Craig–Sulem

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Multi-crest superposition: for k -soliton ICs ($k = 1, 2, 3$), solve each crest independently, superimpose $\eta = \sum_i \eta_i$, $\xi = \sum_i \xi_i$, then recompute the composite $\mathcal{G}$ nonlinearly (not by superimposing individual G_i).

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IC parameters drawn uniformly:

- Amplitudes $a \in [0.05, 0.35]$, directions ± 1 , crests $\in \{1, 2, 3\}$
- Centers sampled with minimum separation 50 units

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Split the dynamics: $\partial_t u = L u + N(u)$ where L is the linear dispersion.

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Integrating factor transform: set $v(t) = e^{-Lt} u(t)$, so $\partial_t v = e^{-Lt} N(e^{Lt} v)$.

Linear flow (exact, spectral), with $\omega(k) = \sqrt{g k \tanh(hk)}$:

$$\hat{\eta}(t) = \cos(\omega t) \hat{\eta}_0 + \frac{\sin(\omega t)}{\omega} g_0 \hat{\xi}_0, \quad \hat{\xi}(t) = \cos(\omega t) \hat{\xi}_0 - \frac{\sin(\omega t)}{\omega} g \hat{\eta}_0.$$

Nonlinear part: advanced with GL2 (2-stage, 4th-order implicit Runge–Kutta):

$$\begin{array}{c|cc} \frac{1}{2} - \frac{\sqrt{3}}{6} & \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{2} + \frac{\sqrt{3}}{6} & \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}$$

4 fixed-point iterations per stage. 8 substeps per Δt . Post-step $\frac{2}{3}$ -rule spectral filtering.

Training Data

Tanaka Soliton Dataset (Shallow Water, $h = 1$)

Generation pipeline:

1. Draw random multi-crest soliton ICs (1–3 crests, $a \in [0.05, 0.35]$)
2. Roll out each IC for $T_{\max} = 200$ with $\Delta t = 0.2$ using GL2-IF
3. Subsample every 20th time step $\Rightarrow$ 200 snapshots per trajectory
4. At each snapshot, store $(\eta, \xi, \mathcal{G}(\eta; h)\xi)$

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Key insight: The Tanaka data covers a rich set of nonlinear wave interactions (soliton collisions, multi-crest dynamics) but only at a single depth.

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Random sea state generation:

1. For each sample, draw spectral parameters:
 - Significant wave height $H_s \sim \text{Unif}[0.1, 0.6]$
 - Peak wavenumber $k_p \sim \text{Unif}[0.06, 0.50]$
 - Bandwidth $\sigma_k/k_p \sim \text{Unif}[1.5, 5.0]$
2. Generate $\hat{\eta}(k)$ from a Gaussian power spectrum, random phases
3. Set $\hat{\xi}(k) = -i \operatorname{sgn}(k) \frac{\omega(k)}{|k|} \hat{\eta}(k)$ (deep-water dispersion)
4. Compute $\mathcal{G}(\eta; h) \xi$ via DNO series (order 6, pad $8\times$)

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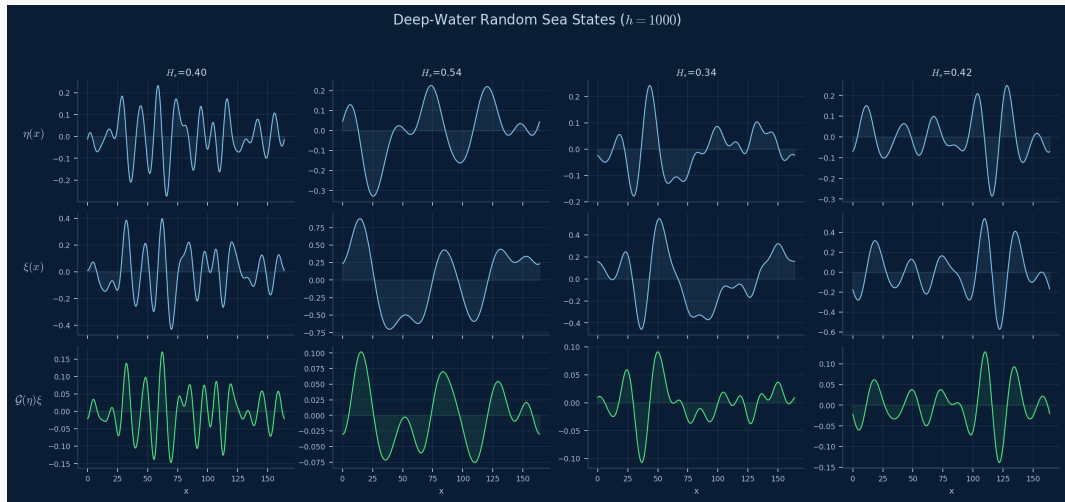
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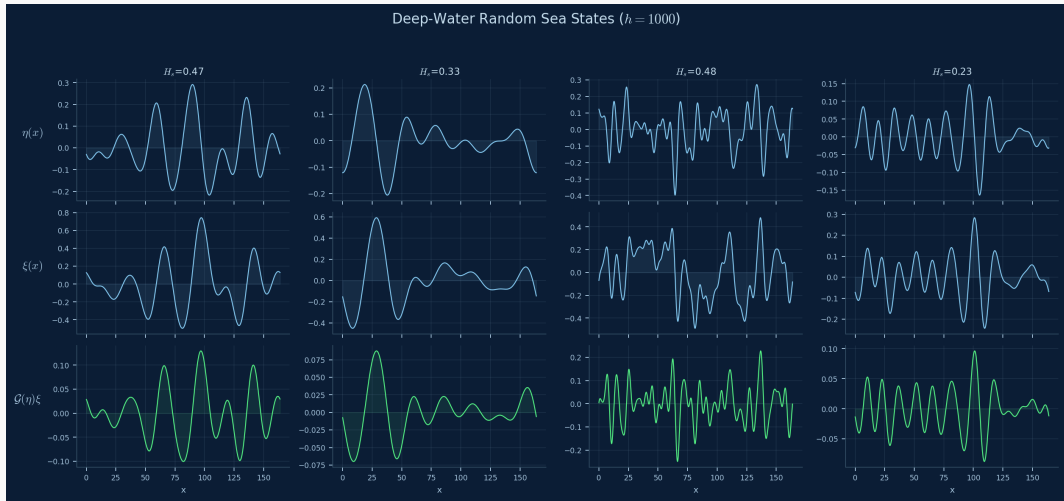
Dataset: 100k samples at $h = 1000$.

Combined training set: 100k Tanaka ($h = 1$) + 100k random sea ($h = 1000$) = 200k total. Each sample labeled with h .

Random Sea Samples ($h = 1000$)



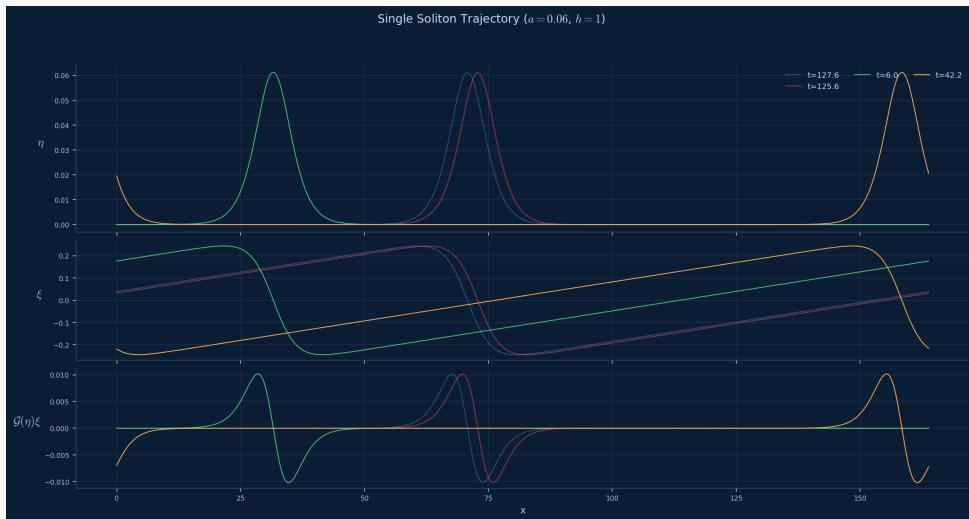
Random Sea Samples ($h = 1000$) (cont.)



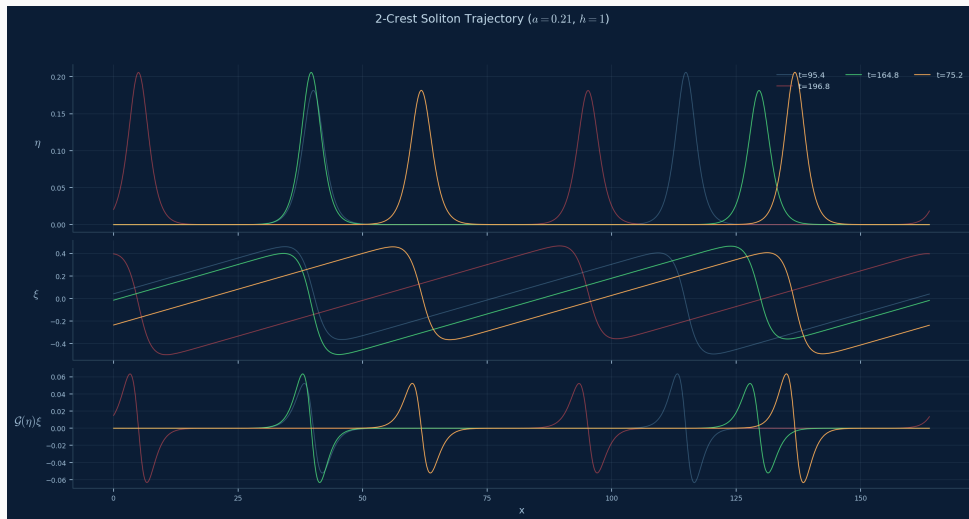
Tanaka Soliton Initial Conditions ($h = 1$)



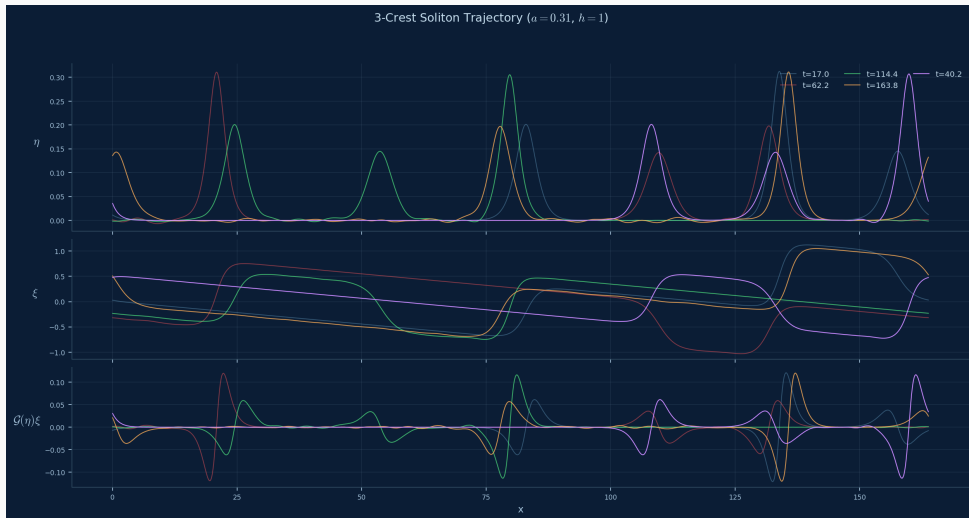
Single Soliton Trajectory ($h = 1$)



2-Crest Soliton Trajectory ($h = 1$)

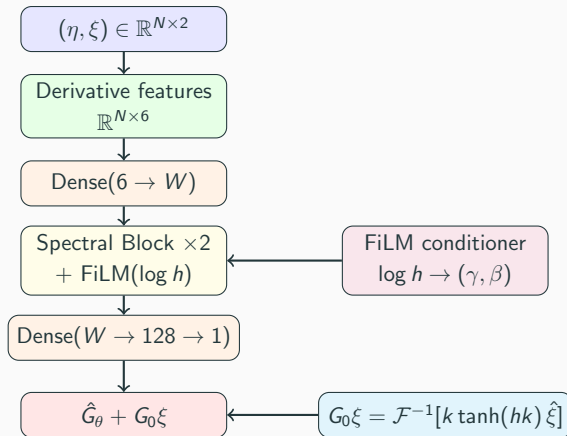


3-Crest Soliton Trajectory ($h = 1$)



FNO Architecture

Architecture Overview



147k parameters. Modes $M = 32$, width $W = 32$, 2 spectral blocks.

1. Linear DNO baseline (residual learning):

$$\hat{\mathcal{G}}_{\theta}(\eta, \xi; h) = \underbrace{G_0(k; h)\xi}_{\text{linear estimate}} + \underbrace{f_{\theta}(\eta, \xi, \log h)}_{\text{learned nonlinear residual}}$$

where $G_0(k; h) = k \tanh(hk)$ is the linear DNO symbol applied as a Fourier multiplier.

The network only learns $\mathcal{G} - G_0\xi$, which is $O(\eta)$.

Key Design Choices

1. Linear DNO baseline (residual learning):

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The network only learns $\mathcal{G} - G_0\xi$, which is $O(\eta)$.

2. FiLM depth conditioning:

$$x^{(\ell)} \leftarrow (1 + \gamma^{(\ell)}) x^{(\ell)} + \beta^{(\ell)}, \quad (\gamma, \beta) = \text{MLP}(\log h).$$

A single network handles all depths via per-layer affine modulation.

3. Spectral convolutions with weight normalization: learned complex weights on the first 32 Fourier modes, with per-mode Frobenius normalization and scalar gains.

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- 4. Derivative features:** finite-difference $\eta_x, \xi_x, \eta_{xx}, \xi_{xx}$ concatenated to input ($2 \rightarrow 6$ channels).

Training Details

Optimizer: AdamW

Learning rate: 5×10^{-4} with cosine decay

Weight decay: 10^{-4}

Batch size: 128

Epochs: 40

Split: 160k train / 20k val / 20k test

Loss: Sobolev loss

$$\mathcal{L} = \text{rel-L}^2(\hat{y}, y) + \text{rel-L}^2_{H^1}(\hat{y}, y),$$

where $\text{rel-L}^2_{H^1}$ weights the Fourier coefficients by $\sqrt{1 + n^2}$.

Normalization: divide features by $|\cdot|_\infty$:

• $n \div 0.75$

Training curve:

Epoch	Train	Val
1	1.078	0.175
10	0.087	0.057
20	0.015	0.013
30	0.008	0.008
37	0.008	0.0074
40	0.007	0.007

Best checkpoint: epoch 37, val loss = 0.0074.

Generalization Results

Motivation: Why Depth Matters

An FNO trained only on Tanaka solitons ($h = 1$) fails to generalize:

- The linear DNO symbol $G_0(k) = k \tanh(hk)$ changes qualitatively with depth
- Deep water ($h \rightarrow \infty$): $G_0(k) \rightarrow |k|$
- Shallow water ($h \rightarrow 0$): $G_0(k) \rightarrow hk^2$
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Solution:

1. Include depth as a FiLM-conditioned input ($\log h$)
2. Provide the linear baseline $G_0\xi$ analytically — the network only learns the nonlinear correction
3. Train on two depth regimes: $h = 1$ (Tanaka) and $h = 1000$ (random sea)

The linear baseline alone achieves $\sim 11\%$ error on the Tanaka data. The learned residual brings this down to $\sim 0.1\%$.

Pointwise Evaluation: Tanaka Solitons ($h = 1$)

Test set: 5,000 Tanaka soliton snapshots (held out from training).

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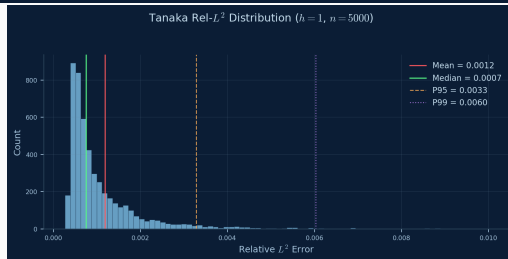
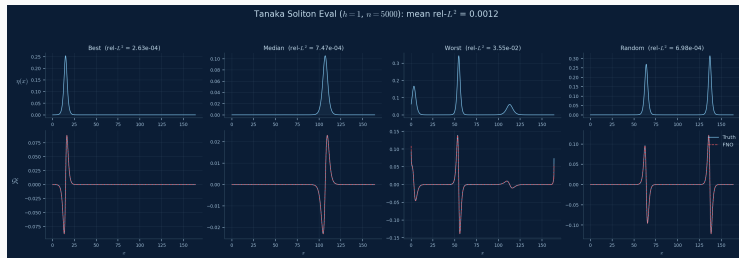
Test set: 5,000 Tanaka soliton snapshots (held out from training).

Metric	Value
Mean rel- L^2	0.0012
Median rel- L^2	0.0007
p_{95} rel- L^2	0.0033
p_{99} rel- L^2	0.0060

Median error is 0.07%. Even the 99th percentile stays below 0.6%.

The FNO captures the full nonlinear DNO across 1–3 crest solitons with amplitudes up to 0.35.

Tanaka Soliton Eval: Samples & Error Distribution



Pointwise Evaluation: DNO Dataset Solitons ($h = 1$)

Test set: 30,030 soliton snapshots from Philippe's Google drive data (not used in training). Amplitudes up to $a = 0.92$, including multi-crest interactions.

Metric	Value
Mean rel- L^2	0.0064
Median rel- L^2	0.0062
Std rel- L^2	0.0098
p_{95} rel- L^2	0.0168
p_{99} rel- L^2	0.0512

Median error stays below 0.7% across 30k independent soliton samples, including amplitudes well beyond the training range ($a \leq 0.35$). The 99th percentile at 5.1% shows graceful degradation on the most extreme cases.

DNO Dataset Soliton Eval: Samples & Error Distribution



Generalization to Stokes Waves

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Deep water ($h = 1000$, 30 samples):

Metric	Value
Mean rel- L^2	4.7×10^{-4}
Median rel- L^2	3.0×10^{-4}

Wavenumbers $n_0 \in \{4, 6, 8, 10, 14, 20\}$,
amplitudes $a_0 \in [0.01, 0.1]$, steepness
 $\varepsilon = k_0 a_0 < 0.3$.

Finite depth ($h = 1$, 28 samples):

Metric	Value
Mean rel- L^2	0.031
Median rel- L^2	0.024

Finite-depth Stokes (Zhao & Liu coefficients). Higher error at large steepness ($\varepsilon > 0.04$).

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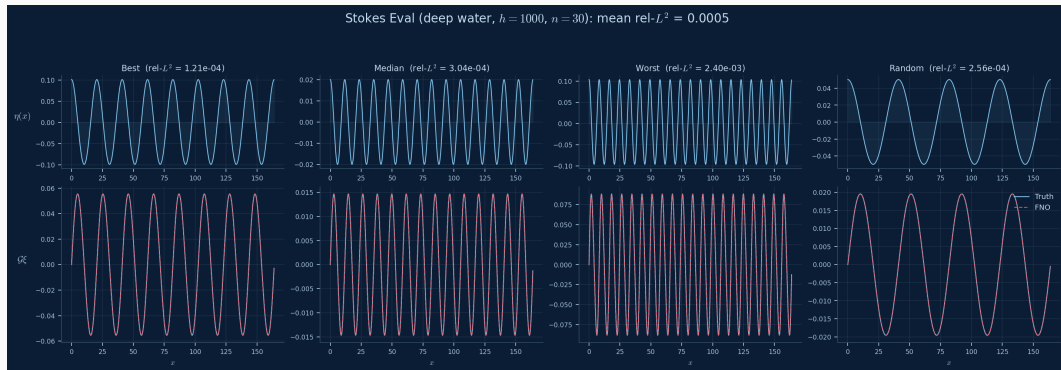
Key: The FNO generalizes from random sea training data to structured Stokes waves at both depths, despite never seeing Stokes profiles during training.

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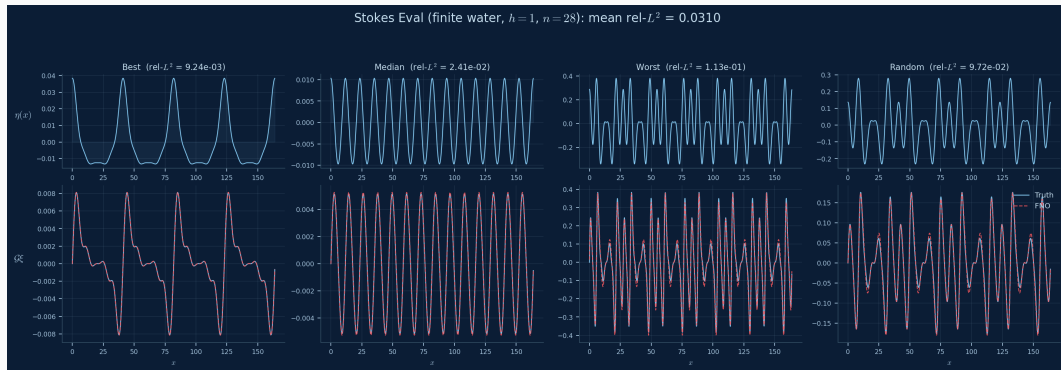
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Stokes Eval: Deep Water ($h = 1000$)



Stokes Eval: Finite Depth ($h = 1$)



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Metric	Value
Mean rel- L^2	0.106
Median rel- L^2	0.093
Std rel- L^2	0.066
P95 rel- L^2	0.229
P99 rel- L^2	0.291

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Why so high? Despite the name “linear,” these samples use ξ evaluated on the exact free surface $z = \eta(x)$ via $\cosh(k_0(\eta + h))$, so the DNO has genuine nonlinear content. The issue is **distribution shift**: these are single-trig-mode cosines at $h = 1$ with steepness up to $k_0 a_0 \approx 2.3$, a profile shape and steepness range absent from training data (solitons + broadband random sea).

Linear Eval: Samples & Error Distribution



Surrogate Rollouts

Surrogate Time Integration

Replace $\mathcal{G}(\eta; h)\xi$ with the FNO in the time integrator:

$$\begin{aligned}\eta_t &= \hat{\mathcal{G}}_\theta(\eta, \xi; h) - G_0 \xi, \\ \xi_t &= -\frac{1}{2}\xi_x^2 + \frac{(\hat{\mathcal{G}}_\theta + \eta_x \xi_x)^2}{2(1 + \eta_x^2)}.\end{aligned}$$

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Speedup: The Craig–Sulem series requires $O(M \cdot N \log(8N))$ per evaluation ($M = 6$, $N = 1024$). The FNO forward pass is dominated by 2 spectral blocks with 32-mode convolutions.

Rollout Accuracy: Short Time ($T = 40$)

Scenario	Depth	$\bar{\eta}$ mean rel-L ²	$\bar{\eta}$ final rel-L ²
Single soliton ($a = 0.40$)	$h = 1$	3.1%	6.1%
Deep Stokes ($n_0 = 10, a_0 = 0.05$)	$h = 1000$	0.21%	0.43%
Finite Stokes ($n_0 = 14, a_0 = 0.10$)	$h = 1$	14.7%	24.6%

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The deep-water Stokes rollout is nearly exact (0.2% mean error over $T = 40$).

The soliton rollout at 3% mean error demonstrates stable propagation of a highly nonlinear wave ($a/h = 0.4$).

Finite-depth Stokes is more challenging due to wave–bottom interaction effects not well-represented in training data.

Rollout Accuracy: Long Time ($T = 400\text{--}600$)

Scenario	$T_{\max}$	$\bar{\eta}$ mean	$\bar{\eta}$ final	$\bar{\xi}$ mean
<i>Soliton overtaking collisions ($h = 1$):</i>				
4:1 amplitude ratio	600	6.4%	20.2%	1.7%
3.5:1 (large amp)	600	4.1%	9.1%	1.7%
<i>Three-soliton interactions ($h = 1$):</i>				
Equal amps (0.38/0.37/0.26)	400	15.0%	42.7%	5.5%
Spread amps (0.38/0.35/0.12)	400	7.4%	17.0%	2.6%

The FNO surrogate tracks **soliton overtaking collisions** through hundreds of time units. Clean overtaking (large amplitude ratio, fast collision) has lower error than slow, prolonged interactions.

Three-soliton dynamics are captured qualitatively, with the spread-amplitude case staying below 7.5% mean error over $T = 400$.

Wall-Clock Performance

Wall-Clock Benchmark (GPU, $N_x = 1024$)

Operation	Method	Mean (ms)	Min (ms)	Speedup
<i>Single $\mathcal{G}$ evaluation (batch = 1):</i>				
	DNO series ($M=6$, pad $8\times$)	3.17	1.22	—
	FNO (2 blocks, 32 modes)	2.26	0.21	$1.4\times$
<i>Single $\mathcal{G}$ evaluation (batch = 64):</i>				
	DNO series	3.55	2.00	—
	FNO	2.19	0.28	$1.6\times$
<i>Full GL2-IF time step (1 sample, 4 iterations):</i>				
	DNO series	21.1	17.5	—
	FNO surrogate	3.2	1.2	$6.6\times$

The raw operator speedup is modest ($1.4\text{--}1.6\times$) since both are FFT-bound at $N = 1024$. But a full GL2-IF step requires ~ 20 recursive DNO evaluations (order-6 series $\times$ 2 stages $\times$ 4 iterations), yielding **$6.6\times$** end-to-end speedup per time step.

Summary

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1. **Solver:** Modified Tanaka ICs + GL2-IF time stepper for the Zakharov–Craig–Sulem system
2. **Data:** 100k Tanaka solitons ($h = 1$) + 100k random sea ($h = 1000$) = 200k training samples
3. **Architecture:** 147k-parameter FNO with linear DNO baseline, FiLM depth conditioning, Sobolev loss
4. **Pointwise accuracy:**
 - Tanaka solitons: 0.12% mean rel- L^2
 - Deep Stokes (OOD): 0.05% mean rel- L^2
 - Finite-depth Stokes (OOD): 3.1% mean rel- L^2
5. **Rollout stability:** Surrogate tracks soliton collisions and 3-wave interactions through $T = 400$ – 600 with 4–15% mean error

Next Steps

- Add intermediate depths ($h = 2, 5, 10, 50$) to the training set for smoother depth interpolation
- Wall-clock benchmarks: DNO series vs. FNO on GPU at various resolutions